Yiran Yuan

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Research Interests

Science of science and citation networks; statistical and computational methods for discovery emergence and knowledge diffusion; large-scale bibliometric data analysis with machine learning.

Education

Southern University of Science and Technology, B.S. in Statistics — Sep. 2023 – Present

- GPA: 3.72/4.0; rank: 10/30; expected graduation: Jul. 2027
- Selected coursework: Probability and Mathematical Statistics, Introduction to Data Science, Data Structures and Financial Applications, Network Science and Computing

University of Pennsylvania, Exchange Student in Statistics — Aug. 2025 – Dec. 2025

- Coursework: Forecasting for Time Series, Introduction to R

Research Experience

Scientific Discovery Birthplaces and Citation Diffusion in Global Knowledge Networks — Apr. 2026 – Present

- Research question: where do major discoveries originate, and how do they diffuse across institutional and geographic citation networks?
- Built discovery-level OpenAlex corpora with entity disambiguation and non-research filtering; pilot audit achieved 86% topical precision on sampled retrieval results.
- Detected discovery-specific burst years with Kleinberg's algorithm (79% aligned with known takeoff periods within $\pm$5 years); identified foundation papers via citation-neighborhood expansion and BERT/citation hybrid ranking (top-10 covered 84% of manually validated canonical works).
- Preliminary finding: citing-city ranks follow a heavy-tailed pattern (Zipf-like, $R^2 = 0.88$ on log-log rank-size fit), suggesting diffusion concentrates in a small set of knowledge centers.
- Journal manuscript in preparation (target submission: late 2026).

Evidence-Oriented Scientific Literature Retrieval and Evaluation — Mar. 2026 – Present

Remote Research Assistant, PI: Shana Stites, University of Pennsylvania

- Contributed to a systematic review of 28 HIV studies on subjective–objective cognitive dissociation; conference poster, AAIC Neuro Next 2026.
- Built a PubMed retrieval pipeline (Sentence-BERT + FAISS) comparing dense, BM25, and hybrid retrieval with Recall@k, MRR, MAP@k, and NDCG for auditable evidence mapping.
- Built a cross-disease evidence map of subjective–objective cognitive dissociation across HIV and Alzheimer's literatures (37 PWH and 48 AD studies extracted; study-level labels grounded in citation-level evidence); poster accepted, Penn CFAR 2026.

Industry Research Experience

Imaging Algorithm Evaluation Intern, OPPO — Summer 2026

- Studied misalignment between objective video-quality measurements and expert subjective severity scores under missing scene labels and heterogeneous score distributions.
- Designed label debiasing corrections and unsupervised scene clustering; trained CNN/Transformer models to predict expert scores while learning latent scene structure.
- Improved objective–subjective consistency from 82.2% to 94.6% for downstream imaging algorithm evaluation.

Selected Projects

User Growth Strategy Optimization with Causal Inference — Nov. 2025 – Jan. 2026

- Estimated user-level CATE with an X-Learner for targeted outreach under an uplift modeling framework.
- Designed a quadratic programming model for budget allocation under channel budget and contact-frequency constraints.

Honors and Awards

MCM/ICM 2025, Successful Participant Award 2025
Academic Excellence Scholarship, SUSTech 2023 – 2025
Outstanding Freshman Scholarship, SUSTech 2023
FLTRP Cup English Speaking/Writing Contest, Bronze Award 2024

Skills

- **Statistics and Network Science:** bibliometrics, citation networks, science of science, burst detection, causal inference, uplift modeling
- **Programming:** Python, R, SQL, Git; Pandas, NumPy, scikit-learn, PyTorch, OpenAlex
- **Machine Learning and NLP:** BERT embeddings, Sentence-BERT, FAISS, retrieval evaluation (Recall@k, MRR, NDCG)
- **Languages:** TOEFL 5.5/6.0 (Listening 6.0, Speaking 6.0), CET-6 631